

Chapter 10 Homework: Endogenous Regressors and Moment-Based Estimation

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Exercise 10.18

Consider the data file `mroz` on working wives. Use the 428 observations on married women who participate in the labor force. We examine the effectiveness of a parent's college education as an instrumental variable.

```
load(url("http://www.principlesofeconometrics.com/poe5/data/rdata/mroz.rdata"))
# Keep only labor force participants
mroz_lf <- subset(mroz, lfp == 1)
cat("Observations:", nrow(mroz_lf), "\n")
```

```
## Observations: 428
```

Part (a): Create MOTHERCOLL and FATHERCOLL

```
mroz_lf$MOTHERCOLL <- as.integer(mroz_lf$mothereduc > 12)
mroz_lf$FATHERCOLL <- as.integer(mroz_lf$fathereduc > 12)

cat("% mothers with some college:",
    round(mean(mroz_lf$MOTHERCOLL) * 100, 2), "%\n")
```

```
## % mothers with some college: 12.15 %
```

```
cat("% fathers with some college:",
    round(mean(mroz_lf$FATHERCOLL) * 100, 2), "%\n")
```

```
## % fathers with some college: 11.68 %
```

Part (b): Correlations between EDUC, MOTHERCOLL, FATHERCOLL

```
cor_mat <- cor(mroz_lf[, c("educ", "MOTHERCOLL", "FATHERCOLL")])
print(round(cor_mat, 4))
```

```
##          educ MOTHERCOLL FATHERCOLL
## educ      1.0000    0.3595    0.3985
## MOTHERCOLL 0.3595    1.0000    0.3546
## FATHERCOLL 0.3985    0.3546    1.0000
```

Interpretation: - MOTHERCOLL and FATHERCOLL should be positively correlated with EDUC (relevant instruments). - They should be reasonably uncorrelated with the wage error (valid instruments).

Part (c): Create COLLSUM; 2SLS with COLLSUM as IV

```
library(AER)

mroz_lf$COLLSUM <- mroz_lf$MOTHERCOLL + mroz_lf$FATHERCOLL

model_iv_sum <- ivreg(log(wage) ~ educ + exper + I(exper^2) |
                      COLLSUM + exper + I(exper^2),
                      data = mroz_lf)
summary(model_iv_sum, diagnostics = TRUE)
```

```
##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | COLLSUM +
##      exper + I(exper^2), data = mroz_lf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.07886 -0.32009  0.03339  0.37479  2.36129
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.2649872   0.3934021  -0.674   0.50095
## educ         0.0867082   0.0308786   2.808   0.00521 **
## exper        0.0427405   0.0132993   3.214   0.00141 **
## I(exper^2)   -0.0008508   0.0003977  -2.139   0.03298 *
##
## Diagnostic tests:
##              df1 df2 statistic p-value
## Weak instruments    1 424   113.357 <2e-16 ***
## Wu-Hausman          1 423    0.576   0.448
## Sargan              0 NA         NA      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6681 on 424 degrees of freedom
## Multiple R-Squared: 0.1525, Adjusted R-squared: 0.1465
## Wald test: 9.634 on 3 and 424 DF, p-value: 3.646e-06
```

```
cat("\n95% CI for EDUC coefficient:\n")
```

```
##
## 95% CI for EDUC coefficient:
```

```
print(confint(model_iv_sum)["educ", ])
```

```
##      2.5 %      97.5 %
## 0.02618724 0.14722911
```

Part (d): First-stage equation; test strength of COLLSUM

```
first_stage <- lm(educ ~ COLLSUM + exper + I(exper^2), data = mroz_lf)
summary(first_stage)
```

```
##
## Call:
## lm(formula = educ ~ COLLSUM + exper + I(exper^2), data = mroz_lf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.2206 -0.3064 -0.2206  0.7425  5.0580
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 11.893672   0.290104  40.998  <2e-16 ***
## COLLSUM      1.964551   0.184518  10.647  <2e-16 ***
## exper        0.049854   0.040108   1.243    0.215
## I(exper^2)  -0.001504   0.001197  -1.256    0.210
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.032 on 424 degrees of freedom
## Multiple R-squared:  0.2148, Adjusted R-squared:  0.2093
## F-statistic: 38.67 on 3 and 424 DF, p-value: < 2.2e-16
```

```
# F-test for COLLSUM in first stage
library(car)
f_test <- linearHypothesis(first_stage, "COLLSUM = 0")
print(f_test)
```

```
##
## Linear hypothesis test:
## COLLSUM = 0
##
## Model 1: restricted model
## Model 2: educ ~ COLLSUM + exper + I(exper^2)
##
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1     425 2219.2
## 2     424 1751.1   1    468.15 113.36 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cat("\nF-statistic for COLLSUM:", round(f_test$F[2], 4), "\n")
```

```
##
## F-statistic for COLLSUM: 113.3573
```

```
cat("Strong instrument (F > 10)?", f_test$F[2] > 10, "\n")
```

```
## Strong instrument (F > 10)? TRUE
```

Part (e): Test

$$H_0 : \beta_{MOTHERCOLL} = \beta_{FATHERCOLL}$$

```
first_stage_sep <- lm(educ ~ MOTHERCOLL + FATHERCOLL + exper + I(exper^2),
                      data = mroz_lf)
summary(first_stage_sep)
```

```
##
## Call:
## lm(formula = educ ~ MOTHERCOLL + FATHERCOLL + exper + I(exper^2),
##     data = mroz_lf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.2152 -0.3056 -0.2152  0.7627  5.0620
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 11.890259   0.290251  40.965 < 2e-16 ***
## MOTHERCOLL   1.749947   0.322347   5.429 9.58e-08 ***
## FATHERCOLL   2.186612   0.329917   6.628 1.04e-10 ***
## exper        0.049149   0.040133   1.225  0.221
## I(exper^2)  -0.001449   0.001199  -1.209  0.227
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.033 on 423 degrees of freedom
## Multiple R-squared:  0.2161, Adjusted R-squared:  0.2086
## F-statistic: 29.15 on 4 and 423 DF,  p-value: < 2.2e-16
```

```
# Test equality of coefficients
eq_test <- linearHypothesis(first_stage_sep, "MOTHERCOLL = FATHERCOLL")
print(eq_test)
```

```
##
## Linear hypothesis test:
## MOTHERCOLL - FATHERCOLL = 0
##
## Model 1: restricted model
## Model 2: educ ~ MOTHERCOLL + FATHERCOLL + exper + I(exper^2)
##
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1     424 1751.1
## 2     423 1748.3  1     2.7258 0.6595 0.4172
```

```
cat("\nReject H0 (coefficients equal) at 5%?", eq_test$`Pr(>F)`[2] < 0.05, "\n")
```

```
##
## Reject H0 (coefficients equal) at 5%? FALSE
```

Part (f): Should we use COLLSUM or separate

MOTHERCOLL and FATHERCOLL?

```
# 2SLS with both MOTHERCOLL and FATHERCOLL separately
model_iv_sep <- ivreg(log(wage) ~ educ + exper + I(exper^2) |
                      MOTHERCOLL + FATHERCOLL + exper + I(exper^2),
                      data = mroz_lf)
summary(model_iv_sep, diagnostics = TRUE)
```

```
##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL +
##       FATHERCOLL + exper + I(exper^2), data = mroz_lf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.07797 -0.32128  0.03418  0.37648  2.36183
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.2790819   0.3922213  -0.712   0.47714
## educ         0.0878477   0.0307808   2.854   0.00453 **
## exper        0.0426761   0.0132950   3.210   0.00143 **
## I(exper^2)   -0.0008486   0.0003976  -2.135   0.03337 *
##
## Diagnostic tests:
##              df1 df2 statistic p-value
## Weak instruments    2 423    56.963 <2e-16 ***
## Wu-Hausman          1 423     0.519   0.472
## Sargan              1  NA     0.238   0.626
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6679 on 424 degrees of freedom
## Multiple R-Squared:  0.153,    Adjusted R-squared:  0.147
## Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06
```

```
cat("\nComparison of EDUC estimates:\n")
```

```
##
## Comparison of EDUC estimates:
```

```
cat("IV (COLLSUM):  ", round(coef(model_iv_sum)["educ"], 4), "\n")
```

```
## IV (COLLSUM):    0.0867
```

```
cat("IV (both IVs):  ", round(coef(model_iv_sep)["educ"], 4), "\n")
```

```
## IV (both IVs):    0.0878
```

```
# Sargan test for over-identification (with two IVs)
summary(model_iv_sep, diagnostics = TRUE)
```

```
##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL +
##       FATHERCOLL + exper + I(exper^2), data = mroz_lf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.07797 -0.32128  0.03418  0.37648  2.36183
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.2790819   0.3922213  -0.712   0.47714
## educ         0.0878477   0.0307808   2.854   0.00453 **
## exper        0.0426761   0.0132950   3.210   0.00143 **
## I(exper^2)   -0.0008486   0.0003976  -2.135   0.03337 *
##
## Diagnostic tests:
##              df1 df2 statistic p-value
## Weak instruments    2 423    56.963 <2e-16 ***
## Wu-Hausman          1 423     0.519   0.472
## Sargan              1  NA     0.238   0.626
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6679 on 424 degrees of freedom
## Multiple R-Squared:  0.153,    Adjusted R-squared:  0.147
## Wald test: 9.724 on 3 and 424 DF,  p-value: 3.224e-06
```

Conclusion: If the test in (e) fails to reject H_0 (coefficients are equal), then using COLLSUM is justified and is equivalent to constraining the two coefficients to be equal. Using both IVs separately over-identifies the model, which can be tested with the Sargan/Hansen J-test.

Exercise 10.20

The CAPM says that the risk premium on security j is related to the market portfolio risk premium:

$$r_j - r_f = \alpha_j + \beta_j(r_m - r_f)$$

We measure the market portfolio using the S&P 500 value-weighted index and the risk-free rate by the 30-day LIBOR monthly rate of return.

```
load(url("http://www.principlesofeconometrics.com/poe5/data/rdata/capm5.rdata"))
cat("Variable names in capm5:", names(capm5), "\n")
```

```
## Variable names in capm5: date time ge ibm ford msft dis xom mkt riskfree
```

```
# Identify the market risk premium column (could be mktrf, market, mkt, etc.)
mkt_col <- intersect(c("mktrf", "market", "mkt", "sp500rp"), names(capm5))[1]
cat("Using market column:", mkt_col, "\n")
```

```
## Using market column: mkt
```

```
# Identify Microsoft return column
msft_col <- intersect(c("msft", "microsoft", "msoft"), names(capm5))[1]
cat("Using msft column:", msft_col, "\n")
```

```
## Using msft column: msft
```

```
# Identify risk-free rate column
rf_col <- intersect(c("riskfree", "rf", "tbill", "t30ret"), names(capm5))[1]
cat("Using risk-free column:", rf_col, "\n")
```

```
## Using risk-free column: riskfree
```

```
# Create standardised columns
capm5$msft_rp <- capm5[[msft_col]] - capm5[[rf_col]]
capm5$mkt_rp <- capm5[[mkt_col]] # mktrf is already excess return in POE5

head(capm5[, c(msft_col, mkt_col, rf_col, "msft_rp", "mkt_rp")])
```

	msft <dbl>	mkt <dbl>	riskfree <dbl>	msft_rp <dbl>	mkt_rp <dbl>
1	0.154255	0.004502	0.00447963	0.14977537	0.004502
2	0.136154	0.073303	0.00447963	0.13167437	0.073303
3	0.056047	0.051301	0.00451929	0.05152771	0.051301
4	0.006983	0.010903	0.00450343	0.00247957	0.010903
5	-0.058946	-0.025748	0.00448757	-0.06343357	-0.025748
6	0.277819	0.031939	0.00449550	0.27332350	0.031939

6 rows

Part (a): OLS estimate of Microsoft's beta

```
model_ols <- lm(msft_rp ~ mkt_rp, data = capm5)
summary(model_ols)
```



```
##
## Call:
## lm(formula = msft_rp ~ mkt_rp, data = capm5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.27671 -0.04764 -0.01104  0.03710  0.35487
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.000369   0.006045   0.061   0.951
## mkt_rp       1.211274   0.122208   9.912 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08062 on 178 degrees of freedom
## Multiple R-squared:  0.3556, Adjusted R-squared:  0.352
## F-statistic: 98.24 on 1 and 178 DF, p-value: < 2.2e-16
```

```
beta_ols <- coef(model_ols)["mkt_rp"]
cat("\nOLS estimate of Microsoft beta:", round(beta_ols, 4), "\n")
```

```
##
## OLS estimate of Microsoft beta: 1.2113
```

```
cat("Classification:",
    ifelse(beta_ols > 1, "Risky (aggressive)", "Relatively safe (defensive)"), "\n")
```

```
## Classification: Risky (aggressive)
```

Part (b): RANK as instrumental variable

```
capm5$RANK <- rank(capm5$mkt_rp)

first_stage <- lm(mkt_rp ~ RANK, data = capm5)
summary(first_stage)
```

```
##
## Call:
## lm(formula = mkt_rp ~ RANK, data = capm5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.109022 -0.005942  0.001608  0.009267  0.027639
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -7.649e-02  2.172e-03  -35.22  <2e-16 ***
## RANK         9.046e-04  2.081e-05   43.47  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01451 on 178 degrees of freedom
## Multiple R-squared:  0.9139, Adjusted R-squared:  0.9134
## F-statistic: 1889 on 1 and 178 DF, p-value: < 2.2e-16
```

```
cat("\nF-statistic for RANK in first stage:",
    round(summary(first_stage)$fstatistic[1], 4), "\n")
```

```
##
## F-statistic for RANK in first stage: 1889.32
```

```
cat("Strong instrument (F > 10)?",
    summary(first_stage)$fstatistic[1] > 10, "\n")
```

```
## Strong instrument (F > 10)? TRUE
```

Assessment: RANK satisfies IV1 (correlated with market return by construction) and IV3 (RANK is exogenous since it's a rank ordering of the market return). The question for IV2 is whether RANK is correlated with the structural error.

Part (c): Hausman test — is OLS biased?

```
capm5$v_hat <- residuals(first_stage)

model_hausman <- lm(msft_rp ~ mkt_rp + v_hat, data = capm5)
summary(model_hausman)
```

```
##
## Call:
## lm(formula = msft_rp ~ mkt_rp + v_hat, data = capm5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.27469 -0.03928 -0.00763  0.03351  0.34402
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5.732e-05  5.989e-03  -0.010   0.9924
## mkt_rp       1.290e+00  1.266e-01  10.195  <2e-16 ***
## v_hat       -9.200e-01  4.314e-01  -2.133   0.0343 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07983 on 177 degrees of freedom
## Multiple R-squared:  0.3718, Adjusted R-squared:  0.3647
## F-statistic: 52.37 on 2 and 177 DF,  p-value: < 2.2e-16
```

```
hausman_t <- coef(summary(model_hausman))["v_hat", "t value"]
cat("\nHausman test t-statistic on v_hat:", round(hausman_t, 4), "\n")
```

```
##
## Hausman test t-statistic on v_hat: -2.1328
```

```
cat("Significant at 5%?", abs(hausman_t) > qt(0.975, df = nrow(capm5) - 3), "\n")
```

```
## Significant at 5%? TRUE
```

Part (d): IV/2SLS estimate of Microsoft's beta

```
library(AER)

model_iv <- ivreg(msft_rp ~ mkt_rp | RANK, data = capm5)
summary(model_iv)
```

```
##
## Call:
## ivreg(formula = msft_rp ~ mkt_rp | RANK, data = capm5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.275605 -0.047708 -0.009281  0.035745  0.352164
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5.732e-05  6.055e-03  -0.009    0.992
## mkt_rp       1.290e+00  1.280e-01  10.083   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08071 on 178 degrees of freedom
## Multiple R-Squared: 0.3541, Adjusted R-squared: 0.3505
## Wald test: 101.7 on 1 and 178 DF, p-value: < 2.2e-16
```

```
beta_iv <- coef(model_iv)["mkt_rp"]

cat("\nOLS beta:", round(beta_ols, 4), "\n")
```

```
##
## OLS beta: 1.2113
```

```
cat("IV beta: ", round(beta_iv, 4), "\n")
```

```
## IV beta: 1.2905
```

```
cat("\nDifference (IV - OLS):", round(beta_iv - beta_ols, 4), "\n")
```

```
##
## Difference (IV - OLS): 0.0792
```

Interpretation: If the market return is measured with error, OLS is biased toward zero (attenuation bias). The IV estimator should be larger in magnitude if measurement error is the problem. Compare the OLS and IV estimates to assess the direction and size of bias.

Part (e): Summary comparison

```
cat("=== Summary: Microsoft Beta Estimates ===\n")
```

```
## === Summary: Microsoft Beta Estimates ===
```

```
cat(sprintf("OLS: %.4f (SE = %.4f)\n",
            beta_ols, coef(summary(model_ols))[2, 2]))
```

```
## OLS: 1.2113 (SE = 0.1222)
```

```
cat(sprintf("IV: %.4f (SE = %.4f)\n",  
            beta_iv, coef(summary(model_iv))[2, 2]))
```

```
## IV: 1.2905 (SE = 0.1280)
```

```
cat("\n")
```

```
cat("If beta > 1: Microsoft is a high-risk stock (amplifies market movements).\n")
```

```
## If beta > 1: Microsoft is a high-risk stock (amplifies market movements).
```

```
cat("If beta < 1: Microsoft is a low-risk stock (less volatile than market).\n")
```

```
## If beta < 1: Microsoft is a low-risk stock (less volatile than market).
```